

LANSERIA, South Africa

WMO: 682635

Lat: 25.939S

Lon: 27.926E

Elev: 1377

StdP: 85.84

Time Zone: 2.00 (E02)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	3.0	-12.2	1.5	15.8	-10.2	1.9	15.9	9.7	15.0	8.4	16.7	1.3	240	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.4	32.8	16.3	31.2	16.3	30.2	16.4	20.2	26.0	19.7	25.4	19.4	25.1	5.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.3	21.2	18.2	15.5	21.0	17.9	15.3	21.0	65.1	26.3	63.4	25.5	61.9	24.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	6.9	DB	-1.0	34.5	1.4	1.2	-2.0	35.4	-2.8	36.1	-3.6	36.7	-4.7	37.6	(4)
(5)			WB	-3.2	21.4	0.9	0.6	-3.8	21.9	-4.4	22.2	-4.9	22.5	-5.6	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	22.4	22.4	21.1	17.8	14.9	12.0	11.7	14.7	18.8	20.8	21.5	22.4
	DBStd	4.52	1.91	1.67	1.82	2.18	2.02	2.00	1.92	2.74	2.78	2.86	2.69	2.40
	HDD10.0	16	0	0	0	0	1	6	7	2	0	0	0	0
	HDD18.3	703	1	1	2	36	107	189	206	116	27	12	6	3
	CDD10.0	3060	384	347	343	233	153	67	60	147	264	334	346	383
	CDD18.3	707	127	114	87	18	1	0	0	3	41	88	102	127
	CDH23.3	6401	848	787	649	216	67	6	7	129	668	994	983	1046
	CDH26.7	1944	254	217	162	28	3	0	0	10	193	362	348	367
Wind	WSAvg	2.8	2.9	2.8	2.6	2.4	2.1	2.2	2.3	2.8	3.2	3.6	3.7	3.3
Precipitation	PrecAvg	602	113	93	86	32	14	3	1	3	12	57	83	104
	PrecMax	1012	243	239	311	107	82	14	7	21	45	157	175	179
	PrecMin	334	23	12	13	1	0	0	0	0	0	6	7	40
	PrecStd	172	56	63	72	27	20	4	2	6	14	40	40	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	32.9	31.8	28.9	27.1	24.2	24.0	28.0	32.0	34.0	34.1	33.9
		MCWB	17.1	17.4	16.0	16.2	12.6	10.8	9.9	12.6	14.0	15.7	15.3	17.4
	2%	DB	31.1	30.9	30.1	27.1	25.1	22.8	22.2	26.1	30.8	32.1	31.9	32.0
		MCWB	17.8	17.6	16.7	15.0	12.1	10.1	9.3	12.0	13.6	14.7	15.9	17.7
	5%	DB	29.2	29.1	28.8	25.9	24.0	21.1	21.1	24.9	29.1	30.8	30.2	30.2
		MCWB	18.3	18.0	16.4	15.0	11.9	9.6	9.3	11.6	13.0	14.8	16.2	17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	28.0	28.0	27.1	24.1	22.9	19.9	19.8	23.1	27.8	28.9	28.9	28.9
		MCWB	18.5	18.2	16.6	14.6	11.9	9.4	9.0	10.5	12.7	14.8	16.2	17.6
	0.4%	WB	21.1	20.6	20.1	18.2	16.3	12.8	12.1	14.1	16.1	18.3	19.5	20.8
		MCDB	27.3	26.7	25.0	25.3	21.3	19.4	18.9	25.3	26.3	26.2	26.9	26.7
	2%	WB	20.4	20.0	19.4	17.4	14.7	11.9	11.1	13.2	15.3	17.5	18.7	19.9
		MCDB	25.9	25.9	24.2	23.8	20.6	18.6	18.3	23.7	25.3	25.6	26.4	26.5
Mean Daily Temperature Range	5% DB	WB	19.8	19.5	18.6	16.6	13.7	11.0	10.2	12.5	14.6	16.8	18.2	19.4
		MCDB	24.9	25.4	23.6	22.4	20.6	18.0	18.1	22.2	24.4	24.9	25.5	25.9
	10%	WB	19.4	19.1	18.1	15.8	13.0	10.3	9.6	11.7	13.9	16.3	17.7	18.8
		MCDB	24.5	25.0	23.0	21.2	19.8	17.3	17.7	20.7	23.5	24.4	24.7	25.1
	Clear-Sky Solar Irradiance	taub	0.398	0.401	0.390	0.341	0.302	0.284	0.286	0.343	0.386	0.419	0.378	0.385
		taud	2.312	2.301	2.312	2.418	2.497	2.521	2.501	2.339	2.236	2.179	2.338	2.339
Ebn at Noon		946	927	907	909	906	905	916	891	896	899	959	960	
Edh at Noon		139	137	129	106	89	82	87	113	137	154	135	136	
All-Sky Solar Radiation	RadAvg	6.64	6.28	5.57	4.77	4.37	3.97	4.31	5.08	6.05	6.49	6.75	6.80	
	RadStd	0.70	0.57	0.51	0.42	0.22	0.20	0.11	0.23	0.27	0.44	0.62	0.41	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87	

Nomenclature: See separate page